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Balancing formal and informal success factors perceived by supply chain stakeholders: A study of woody biomass energy systems in Japan

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ABSTRACT

Small-scale woody biomass energy systems have an inherent ability to aid in emissions reduction while stimulating local economies and, as collective energy systems, are strongly connected to supply chain design based on local conditions and stakeholder integration. Despite an abundance of forest area alongside the promotion of biomass in energy policies, however, woody biomass utilization still remains low in Japan. The woody biomass supply chain, considered as a socio-technical system, involves a complex, cross-sectoral stakeholder network in which inter-organizational dynamics necessitates well-organized management based on an understanding of formal factors such as technology, as well as informal factors such as social relations and culture. In this paper, success factor perceptions from across the woody biomass supply chain are investigated based on semi-structured interviews with four stakeholders in the Kyushu region of Japan. Identified success factors here are: 1) respect of values & traditions, 2) transportation infrastructure, 3) business model integration, 4) relationship & trust, 5) local vitalization and 6) biomass quality control. A convergence as well as divergence of perceptions are observed, involving both formal and informal dimensions. Aiming to balance perceptions and to enable long-term success of woody biomass in Japan, a series of policy implications are drawn, including cross-ministerial integration, knowledge building on wood logistics, forest certification, local coordinators, biomass quality control standards and a feed-in-tariff for heat. This paper suggests a new arena of policy-making based on the importance of considering both informal and formal dimensions in energy policy.
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1. Introduction

Japan is among the world's top importers of fossil fuels. Being the world's third largest economy and the OECDs' second largest electricity market (Plessis, 2015), alongside being practically barren of conventional energy sources, Japan is in a form of policy deadlock" in its energy and economic development (Portugal-Pereira and Esteban, 2014). However, following the Great East Japan Earthquake in 2011 and the resulting nuclear shutdowns, there was a form of mindset shift in Japan, both in the private and public sector. Renewable energy is increasingly considered vital in the future energy mix, and the number of renewable energy projects has risen rapidly. The government targets a renewable energy share of 25–35% of total power generation by 2030 (IGEL, 2013). An indicator of this mindset change and catalyst of increased shares of renewable energy is the feed-in-tariff (FIT) system established in 2012, superseding the renewable portfolio standards scheme and revising the existing purchasing scheme for PV electricity (IEA, 2016). In addition, the power and gas retail market deregulations of 2016 and 2017 and the legal unbundling of power transmission and distribution and of gas pipelines in 2020 and 2022 are further developments that have potential to drastically alter the energy landscape of Japan. Deregulation of markets tend to lead to accelerated innovation and development, also applicable to the energy sector (Markard and Truffler, 2006).

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Japan's FITs for biomass are amongst the highest worldwide. April 2012–2014, the FIT was 32 JPY/kWh (0.28 USD/kWh, 2017-07-03), for power generated utilizing woody biomass from domestic forest thinning (Izumi et al., Personal communication). From April 2015 and also following the FIT review of April 2017, the highest tariff, at 40 JPY/kWh (0.36 USD/kWh, 2017-07-03), has been granted for power capacities below 2 MW (METI, 2017). The FITs are based on costs and potential revenue involved in creating and running the plant (Izumi et al., Personal communication). Japan's land area is 70% forest (World Bank, 2015), while the energy system's incorporation of woody biomass is low, even following the high FITs (IEEJ, 2014). Many trees are mature in large forests planted following WWII, incurring a high quantity of unutilized forest biomass, with annual residues reaching 20 million m^3 (Yanagida, 2015). At the same time, the forest industry in Japan is unproductive. In the cost structure of running woody biomass electricity generation, feedstock comprises a high share of just under 77% (Yanagida et al., 2015). As highlighted by Plessis (2015), Japan has the fifth largest total potential biomass market in the world, and governmental targets are considerably low when compared to this potential. The target is to increase biomass power generation to 32.8 TWh by 2030, incurring a biomass share of approximately 4%, or 20% of all renewable energy (Plessis, 2015). The share of biomass was 1.5% of total power generation in 2014 (ISEP, 2015).

As explained by Kuboyama & Yanagida (Personal communication), the Ministry of Economy, Trade and Industry (METI) created the FITs in 2012 to encourage the expansion of renewable energy, and the certified capacity of biomass as of October 2015 was 2.7 GW while the actual running capacity was a mere 0.344 GW, or just under 13% of the certified power. Of the running capacity of biomass, domestic forest-based biomass accounts for approximately 0.125 GW, or 36% (Kuboyama & Yanagida, Personal communication). As highlighted by Takanobu (2016), in July 2016, the total certified capacity under the FITs grew to 3.76 GW, of which 0.436 GW accounted for domestic woody biomass from forest thinnings, and 3.32 GW for "ordinary wood", which also includes imported biomass. Domestic wood chip demand for the 0.436 GW capacity can be estimated to 5.56 million tons, the price of fuel from forest thinnings was at 7800 JPY/t (68.9 USD/t, 2017-07-04) in 2015 in contrast to the price of 12,000 JPY/t (106 USD/t, 2017-07-04) as projected for 2015 by METI at the onset of the FIT scheme (Takanobu, 2016). In addition, it is forecasted that the FIT scheme will lead to an annual demand of biomass from domestic forest thinnings of 5.56 million tons by 2030, while the Cabinet's Basic Plan for Forest and Forestry incorporates a potential domestic supply of 5.2 million tons by 2025 (Takanobu, 2016). In all, this indicates a significant imbalance between the potential supply and demand of woody biomass in Japan, alongside discrepancies in certified versus installed power. Despite seemingly fruitful policy conditions, especially for small scale woody biomass, as well as an abundance of forest in Japan, there are challenges being faced which call for further investigation. The purpose of this study is to develop a holistic understanding of the woody biomass supply chain in Japan in order to identify key challenges and success factors from which implications for energy policy can be derived.

2. Woody biomass supply chains: theoretical background

There is inherent potential in collective bioenergy systems to reduce emissions and stimulate local economies (Suntana et al., 2012; DeWit, 2015). Hence, small-scale woody biomass energy systems are investigated in this paper, due to abilities to integrate such systems with local supplies and demands and aid in local revitalization strategies in Japan, important in tackling national issues, such as aging societies. In the context of this study, it is important to define the concept of "small-scale" distributed energy. Pepermans et al. (2003) and Ackermann et al. (2001) describe distributed generation in terms of connection and location, being connected directly to the grid on the customer side of the meter, rather than scales of capacity. Similarly, the small-scale woody biomass concept of this study involves a supply chain based on domestic forest that can be integrated with local industry and demands.

The productivity of forestry in Japan is low, despite high potential and abundant forest. As indicated by Weiss et al. (2011), innovation is key in driving economic growth and enabling competitiveness. As forest-based value chains have several pathways, such as for timber construction and as renewable energy, there is high innovation potential for forestry value pathways. In contrast to solar PV, integration of biomass in local areas is essential in order to ensure stable demand and supply levels (Hisashi et al., Personal communication). Establishment of new biomass operations in an area, both including fuel supply and electricity and/or heat generation, will enhance the competitiveness of current local industries and stimulate new businesses, such as pellet manufacturing (Yagi and Nakata, 2011). Kishita et al. (2017) highlight the ability of woody biomass business to contribute to emissions reduction, energy security and rural economy revitalization. Similarly, Malico et al. (2016) highlight that forest ecosystems can provide many benefits to society, including socioeconomic development, reduced risk of forest fires, biodiversity preservation and biomass for heating and power, and are hence important to manage effectively in order to derive this value. In addition, Leban et al. (2016) discuss a need to engage in management for forest multifunctionality, in an integrative approach through which woody biomass is a by-product of other forest products and services, in a contemporary forest management paradigm.

The Japanese government recognizes woody biomass as a potential means of revitalizing local societies. In the 4th Strategic Energy Plan of 2014, moreover, the importance of producing both power and heat from biomass is highlighted, based on economic efficiency and local conditions in the region at hand (METI, 2014). Heat utilization is especially important for small-scale plants, enabling system efficiencies of 80–90% as opposed to 40% with only power production (Beith, 2011). However, there is no financial incentive given for renewable heating in Japan (METI, 2014). Key pre-requisites of heat utilization from biomass in cogeneration efforts include high levels of local industry integration, the development of a heat distribution network and identification of local heat demand (Hisashi et al., Personal communication). District heating infrastructure has been developed to a limited extent in Japan today (Takanobu, 2016). It can thereby be suggested that industrial heating through co-location with heat demands is the main opportunity for biomass heat utilization. Moreover, central government initiatives, such as forest thinning and "Biomass town" concepts aiming to create bioenergy ecosystems in local towns in Japan have not created any large-scale effects in the country. Yoshida et al. (2014) point out that during the evaluation of the Japanese government's "Biomass town" initiatives, one of the main reasons of failure was lacking rural infrastructure leading to difficulties in collection and transportation of wood from the forest. There are clear weaknesses in the woody biomass supply chain, and in connecting the downstream and upstream supply chain stakeholders.

2.1. Supply chain innovation

In the area of forestry, Gain and Watanabe (2014) highlight that main issues being tackled in Japan in contrast to global standards and progress is its mountainous terrain alongside fragmented,

small-scale forest ownership resulting in a lack of wide-scale efforts, such as thinning, which are in actuality highly promoted by the Japanese government. A lack of effective documentation and forest certification systems has also contributed to unproductive forestry (Gain and Watanabe, 2014). In addition, Ooba et al. (2015) discuss that the affordability of Japanese forest products is lower than that of imported options due to poor management of Japanese forest, and that a focus on imported timber rather than the improvement of local forest management is in turn leading to further degradation of forest ecosystems. The low utilization of locally sourced biomass is also in part attributable to a lack of homogeneity in the quality of woody biomass obtainable from local supply chains, indicating the importance of more effective supply chain management (Ooba et al., 2015). It can therefore be suggested that there is a lack of sourcing and development in the upstream woody biomass supply chain, despite a multitude of forest material in Japan. Integrating energy flows of different businesses in creating collective energy systems, according to Johansen and Røyrvik (2014), majorly involves finding synergies in regional clusters through stakeholder management alongside supply chain design based on local conditions and stakeholders, an important consideration at the onset of planning. According to Hartmann (2014), innovation can encompass several players from one or more organizations in creating new value, whether related to technology, processes or organizational structure. Seeing that the woody biomass supply chain is affected by multiple cross-sectoral stakeholders, and also facing a series of challenges in the Japanese context, the potential of innovation to enable its development is of interest to investigate further.

As indicated by Hugos (2011), supply chain innovation is connected to business model integration and flexibility across the supply chain, rather than focusing on optimizing throughput efficiency, as indicated in traditional supply chain theory. According to Keefe et al. (2014), woody biomass energy systems are often highly dependent on logistics, considering tendencies towards high costs of harvesting, collection, processing, storage and transportation. Aguilar (2014) further highlights that both developing the supply chain design and expanding forestry skills to support it can contribute to woody biomass success. For instance, by introducing centralized chipping earlier on upstream, the overall supply chain efficiency can be increased. Similarly, de la Fuente et al. (2017) find that forest biomass considerations and processing carried out further upstream in the supply chain lead to lower emissions and higher productivity. Removing stem residue (tops, branches and non-merchantable logs) early on in the supply chain can ensure separation from merchantable stems which will be processed for timber and pulp production. This entails that, as indicated by de la Fuente et al. (2017), a higher volume of woody biomass material for energy purposes can be obtained from the forest and the overall value created in the supply chain increased. Aguilar (2014) further points out the lack of literature concerning the social complexities of wood, utilizing the term "social availability", which incorporates both physical raw material as well as harvesting constraints, competition on the wood market as well as the objectives of forest owners. This is also necessary to be included in policy-making processes. White and Bourne (2007) highlight the importance of management science in the complex arena of policy-making in order to incorporate values among stakeholders for integrated solutions.

As such, as highlighted by Hugos (2011), "coordination is more powerful than control" and the ability to maintain profitability in the long-term depends on abilities to manage supply chain networks. There lies a necessity in stakeholder coordination for innovation in integrated energy systems (Johansen and Røyrvik, 2014). Hugos (2011) further highlights the significance of a "mindful approach" in supply chain coordinative efforts, meaning that mindfulness at an individual stakeholder level enables openness, attentiveness to change, sensitivity to different contexts and understanding of multiple perspectives. Mindfulness hence aids in innovation as it opens several angles to tackle a problem and enables adaptivity and the creation of new value. In addition, as highlighted by Yeniyurt et al. (2014), working relations and interdependence affects the attitudes of stakeholders in the supply chain towards co-innovation and associated behaviour. It is further shown that the involvement of suppliers, i.e. upstream supply chain actors, in downstream developments increases the performance of the entire supply chain, among all parties. Energy system infrastructure is tightly integrated with society and dependent on different social networks, and can therefore be referred to as a socio-technical system. By doing so, the interactions between the social and technical system aspects can be taken into account, thereby creating an understanding of the supply chain reality, including its challenges.

2.2. The importance of stakeholder management for supply chain innovation

In accordance with Shiroyama et al. (2015), if feasibility is defined as the prospective of consensus among stakeholders, then in systems with diverse stakeholders, such as the case of woody biomass, it is vital to conduct analyses of complex stakeholder networks in order to find interests and challenges therein. This overview can thereby enable more robust decision-making for improved collaboration from a system perspective. The management of transitioning sociotechnical systems, such as collective woody biomass energy systems, necessitates new forms of governance (Loorbach, 2007). By taking a multiple-stakeholder perspective in supply chain management, this study aims to present new dimensions for policy creation related to woody biomass.

As highlighted by Bolton and Foxon (2015), the integration of social dynamics in the analysis of technical systems, which thereafter can be more accurately referred to as sociotechnical systems, creates room for new academic discourse related to the policy arena. Cooperation across the supply chain can in itself be seen as a form of governance-related innovation, and in addition, a catalyst for innovation by promoting communication and knowledge-exchange. This can be a main factor in closing a perceptions gap across supply chain stakeholders. The potential of innovation in supply chain management is therefore interesting to investigate further, and the abilities of policy to enable this. As highlighted by Kato et al. (2013), especially when multiple viewpoints are involved, it is a main challenge for policy makers to comprehend the full spectrum of opinions and the "problem structure" surrounding which new policies are to be created. Misunderstanding here may lead to a stalemate in policy-driven change (Kato et al., 2013). Problem structuring methods is especially useful in the presence of multiple actors, perspectives, conflicts, intangibles and uncertainties, as it involves understanding the perceptions of all stakeholders, as well as convergences and divergences among them, which is of upmost importance in information creation for public policy development (Kato et al., 2013; Mingers and Rosenhead, 2004; Cronin et al., 2014). Complex problems involving multiple stakeholder perceptions are often referred to as "wicked problems" in policy literature (Cronin et al., 2014). In tackling such wicked problems, Kato et al. (2013) emphasize that visualizing the perceptions of stakeholders is highly beneficial in policy-making as they react differently to policies presented in different frames. Hence it is important to understand the objectives and perceptions of individual stakeholders in order to understand the system in its entirety (Kato et al., 2013).

In the area of policy-making, White and Bourne (2007) discuss that there is a growing necessity of management science and operational research, as well as a greater awareness of complexities involved in wicked problems. A key factor in incorporating multiple stakeholder perceptions in policy-making is the emphasis on dissimilarities as well as convergences of values, which would require the collective engagement of multiple agencies, from communities to the central government, for a top-down and bottom-up approach (White and Bourne, 2007). By creating an overview of perceptions, and understanding the convergences and divergences among different stakeholders therein, there is potential to manage conflict and create new understanding in underpinning areas previously concealed in the mental models of participants. By deconstructing these mental frames and the balance or imbalances among them, stakeholders' relations can be comprehended and managed more effectively. Increased engagement which can be a result of this escalated understanding can aid in enabling innovation (White and Bourne, 2007). Similarly highlighted by Hugos (2011), Weiss et al. (2011) and Loorbach (2007), innovation is strongly linked to engagement and coordination throughout the supply chain in finding synergies among stakeholders. Johansen and Røyrvik (2014) emphasize this specifically in the context of collective energy systems, as is the case of small-scale woody biomass. The importance of understanding and managing perceptions specifically across the woody biomass supply chain is also highlighted by Gold (2011), indicating the high impact of citizen perceptions, Aguilar (2014), implying that challenges are often attributable to different world views related to woody biomass, and Keefe et al. (2014), stating that different objectives of stakeholders across the supply chain need to be coordinated. Yoshida et al. (2014) also highlight the need to foster information symmetry and a common vision in woody biomass supply chains due to the complex stakeholder network. There are many factors mentioned in previous research, indicating the significance of stakeholder management and social dynamics for successful biomass supply chains. However, it is not clear which factors are the most important and how to manage them.

2.3. Importance of both formal & informal factors

Among stakeholders in a social network, as discussed by Bolman and Deal (2008), there are different "frames" or mental models leading to different theoretical and conceptual foundations. This may lead to information and perception asymmetries in the supply chain, leading the project failure. Hence, as also implied by Trapero et al. (2012) and Badakhshan et al. (2015), information symmetry and a common vision is vital in complex supply chain success, also applicable to woody biomass supply chains with convoluted networks of stakeholders (Yoshida et al., 2014). There is a need of a multi-frame approach for complex systems, in order to understand sub-aspects and thereby understand the system as a whole (Thompson, 1967). With regard to energy systems, economic theory has significantly influenced governance and market design. However, as discussed by Bolton and Foxon (2015), solely basing energy governance on technology or economics provides a weak base upon which low-carbon investments can be made in the necessary timescale. Socio-technical shifts, including privatization and deregulation as seen in Japan's energy market reforms, create new scope for academic discourse, such as in the realm of policy development (Bolton and Foxon, 2015). In this development, Bolton and Foxon (2015) highlight that rather than solely focusing on technical intervention in energy systems it is vital to take a systems approach. With this approach, in understanding the system as a whole it useful to analyze supply chain success from both formal and informal dimensions. As explained by Schiefloe (2016), formal qualities involve structure and technology, while informal qualities involve social relations and networks, interaction and culture. With these two perspectives, the supply chain can be understood through system-oriented analysis, based on social constructivism theory, appropriate when analyzing sociotechnical systems (Rolstadås et al., 2014).

There is an availability of literature covering individual elements and technologies in the woody biomass supply chain, such as sustainable forest management (Fabusoro et al., 2014) in the upstream and biomass fluidized bed boiler technologies (Verma, 2014) in the downstream. Several authors show the importance of managing the moisture content of wood throughout the supply chain. Kishita et al. (2017) show the importance to utilize forest wood residue for power generation in order to maintain a sustainable biomass business over time integrated with waste management practices. The low moisture content of this residue, at 10.7%, is also beneficial for use in power production. Additionally, Greene et al. (2014) show that a focus on moisture content reduction can aid in improving biomass logistics, and Malek et al. (2017) highlight the linear relationship between a higher moisture content and a decreased lower heating value, which decreases energy conversion efficiency and causes fuel costs to rise. In all, there is an availability of knowledge on individual elements in the woody biomass supply chain. In Japan, however, an abundance of forest alongside technology availability is yet to lead to higher levels of woody biomass utilization. As Aguilar (2014) and Keefe et al. (2014) indicate, understanding and innovating the supply chains of woody biomass is key for successful utilization in energy systems, due to dependencies on logistics and a stable flow from the upstream to downstream. BR&D (2014) further emphasize that most research and development of bioenergy systems has focused on a single component in the supply chain, rather than taking a holistic approach involving multiple factors. An example of a multi-faceted research approach can be seen in a study on biomass supply chains in Northeastern China carried out by Wang and Watanabe (2016). This study utilizes a multi-dimensional model considering socio-demographic, policy-related, economic and trust factors to analyze farmer risk perceptions and it is found that farmers require economic incentives on the short term to cooperate with middlemen connecting them to bio-energy plants, while on the long term, trust-building and improved educational levels are needed in order to decrease farmer risk perceptions related to the biomass supply chain (Wang and Watanabe, 2016). In the integrated approach needed in biomass research, various parameters are involved, from technology to socioeconomics, concerning each stakeholder as well as corresponding inter-relationships. This is in line with the need to understand inter-organizational dynamics in green supply chains, as highlighted by Sarkis et al. (2011). By taking an integrated approach, a holistically comprehensive understanding of forest product pathways, including formal and informal dimensions, is enabled.

3. Material and methods

In order to illustrate the importance of both formal and informal factors in woody biomass supply chain management as well as elaborate upon the most crucial factors among them, a series of four case studies were carried out in Japan. Eisenhardt (1989) describes the process of theory induction based on case studies to be especially valuable in new research domains, with the ability to create insight by juxtapositioning potentially contradictory findings and adding multiple perspectives. Similarly, Yin (2009) discusses that for a focus on contemporary phenomena with real-life contexts and many variables, case studies are beneficial. The aim of a case study is to provide description, test theory or generate theory

Table 1
Overview of the roles of companies A-D in the biomass supply chain (X = activity; (X) = planned activity).

	Forest Owner	Forest industry	Biofuel producer	Electricity/heat producer
Company A	X	X	(X)	
Company B	X	X	X	X
Company C			X	
Company D			X	X

(Eisenhardt, 1989). This is in line with the aim of this paper to explore a new research area by describing the perceived success factors in the woody biomass supply chain in Japan and generating learnings on the formal and informal factors therein. Firstly, literature review was carried out, as well as semi-structured interviews with 24 experts in Japan presented in the Acknowledgements of this paper. These interviewees have expertise in various areas of woody biomass, from fields of policy, industry and research. Experts from the same organization were interviewed in groups. The interviews incorporated general questions on success factors and challenges of the woody biomass supply chain in Japan, as well as specific questions relevant to the expertise of the interviewees. These interviews and a broad literature review enabled an overview of elements in the supply chain which may be identified as success factors, such as transportation and biomass quality. Eisenhardt (1989) discusses that this form of priori specification of constructs is helpful for more accurate construct measurement in case studies, and that when similar constructs emerge in the data collected there is a strong triangulated ground of research.

Secondly, knowledge gained was used as a basis for the case studies of this paper. The case studies were carried out through interviews with four companies, denoted as Company A, B, C and D. These companies work across the woody biomass supply chain in the Kyushu region, the southernmost island in Japan. These cases were framed in the context of rural, small-scale, locally sourced woody biomass, which could in turn also be found in other places. Regardless of their relations, the companies were selected based on their roles in the supply chain, as seen in Table 1, in order to create an overview of the entire supply chain. The boundaries of the supply chain and the roles therein adopted in this paper can be seen clarified in Fig. 1. Eisenhardt (1989) and Eisenhardt and Graebner (2007) discuss that cases should be selected firstly based on a specified population in order to reduce extraneous variation and thereafter on theoretically useful conceptual categories. In line with this, the population of the Kyushu region and specific representative stakeholder roles therein were selected for this paper. Kyushu is one of the regions in Japan with the highest potential for woody biomass, both in scenarios with current and more advanced harvesting technology (Kamimura et al., 2012). It was hence selected as the targeted region of this study.

This case studies were based on semi-structured interviews with open-ended questions, aiming to find key factors and challenges of successful woody biomass supply chains as identified by the stakeholders. The interview design used with Companies A to D were focused based on the role(s) the firms play in the upstream to downstream supply chain. In addition, the interviews focused on overall challenges and opportunities perceived in the supply chain, in order to build an improved understanding of the multiple.

Perspectives across it. All interviews were carried out with translation between Japanese and English, except with Company A of which all interviewees were fluent in English. The process of translation potentially inhibited discussions and led to a loss of information. Interviewer bias was avoided with two investigators which identified the perceived success factors and thereafter linked them to previously identified success factors in the expert interviews and literature review. Stakeholders' perceived supply chain success factors were identified based on emphasis given by them during the interviews, ranked as strong, moderate or weak by the investigators. Based on this, a comparative analysis was created in tabular form in order to identify convergences and divergences of these perceptions. Tables enable summarization and visualization of findings (Eisenhardt, 1989), especially useful with multiple cases and for increasing testability (Eisenhardt and Graebner, 2007). Indications from this analysis were thereafter utilized to create policy recommendations for woody biomass in Japan. The roles of each company, as seen in Table 1, are described as follows.

3.1. Company A

Company A, a family-run business, is the largest forest-owner in the Oita prefecture on the Kyushu Island, and among the largest in Kyushu, with forest area inherited throughout generations. Company A is also active in the forest industry, selling products such as construction wood. The company has developed road networks in its forest, using machinery, in order to enable the wood harvesting and collection process on mountainous terrain. Company A is also running feasibility projects involving mobile wood-chippers, i.e. chippers on top of harvesting trucks, looking into the potential of entering the woodchip market through enhanced forest residue collection processes. The company highly emphasizes the importance of innovating wood logistics in its mountainous terrain.

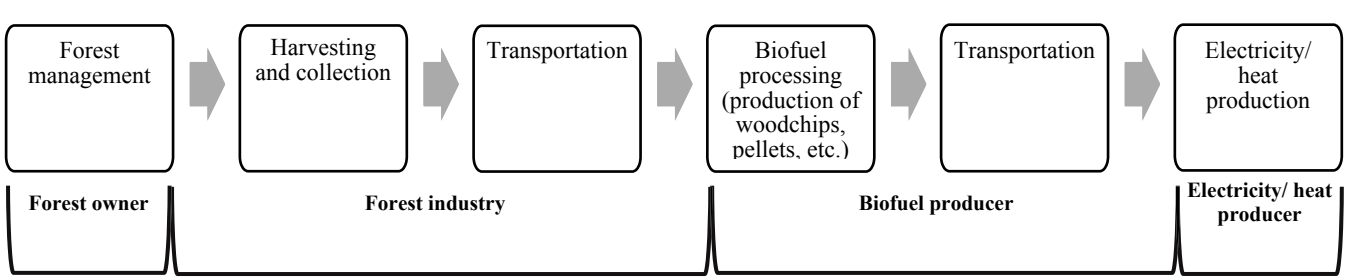

Fig. 1. Framework of woody biomass supply chain and roles therein (based on Lautala et al., 2015).

3.2. Company B

Company B is active throughout the woody biomass supply chain. The company is a forest owner in the Miyazaki prefecture, also sourcing from other forest owners. Company B sells a diversity of forest products, such as construction timber, chopsticks and biofuels, such as briquettes. Similarly to Company A, Company B forest is at large inherited throughout generations. Hence, emphasis is placed on respect of the forest, and the importance of deriving the most value from its trees. Hence, woody biofuel is given as the final step of the value capture from the forest, for which residues from the manufacturing of other forest products are preferably used. For example, the residues of chopstick manufacturing are used to make briquettes. Company B utilizes own woodchips for power and heat generation, mainly for utilization in the company's operations. Excess power is sold to the grid.

3.3. Company C

Company C is solely a biofuel producer, specifically of wood pellets, located in the Miyazaki prefecture, sourcing all of its lumber from local forest industry players. The entire wood pellets supply of company C is sold to a single large scale utility in Japan for co-firing with coal. A local forest association in Miyazaki, in addition to the power utility, is a shareholder in Company C. A main reason for this is to increase the credibility of Company C among local forest industry actors and enable sustainable relationships and trust, in order to secure a stable wood supply.

3.4. Company D

Company D is an electricity generator at a scale of 5.7 MW. Electricity is sold to the grid, and waste heat from power production is shared for no charge to local farmers, utilized for strawberry cultivation in greenhouses for which Company D also provides land free of charge. Giving back to local communities is given as a key part of Company D's business model. Company D also has strict routines for the testing of biomass moisture as well as categorization of woodchips in different moisture intervals for moisture level regulation inside the boiler of the power plant, in real-time. The owner of Company D also owns a company producing and selling woodchips in Kyushu. Hence, a stable supply and quality of woodchips can more easily be secured.

4. Results and discussion

Based on the interviews of this study, six success factors which stakeholders emphasized as vital across the woody biomass supply chain were identified, as presented in Table 2. The success factors are: 1) Respect of values & traditions, 2) Transportation infrastructure, 3) Business model integration, 4) Relationship & trust, 5) Local vitalization and 6) Biomass quality control. Table 2 presents an overview of success factor perceptions among the stakeholders of the woody biomass supply chain, indicating both a convergence and divergence of perceptions.

All six factors are involved across the supply chain, but are emphasized differently from upstream to downstream stakeholders. As follows is an overview of the convergence and divergence of perceptions of woody biomass supply chain success factors, followed by policy indications in order to promote locally sourced woody biomass in Japan by leveraging convergences and managing divergences.

4.1. Convergence of perceptions

Finding that most actors agree on several success factors identified in Table 2, there is a foundation for strategic development in these areas. However, there are slightly different views on which approaches to use. Reaching a common understanding on the most efficient and mutually beneficial supply chain design to enable the success factors surrounding which there is a convergence may facilitate knowledge-sharing and join investments in required technology and equipment. Companies A, B and D generally emphasize the importance of the success factors 2) transportation infrastructure, 3) business model integration, 4) relationship and trust and 5) local vitalization. Company C does not emphasize 3) business model integration and 5) local vitalization, as its business model is focused on supplying wood pellets to one large-scale utility. These factors are shown to be given higher emphasis in the context of small-scale woody biomass, for which it is beneficial for business models to be integrated with local communities, demand and supplies. This can be seen, for instance, in Company D's strong emphasis on local community vitalization by enabling farmers through waste heat channels. The convergence of success factors is discussed as follows.

4.1.1. Transportation infrastructure

Pertaining to success factor 2), transportation infrastructure, all companies indicate infrastructure for the collection and transportation of wood out of the forest as a main success factor in the woody biomass supply chain, especially when considering the mountainous terrain of Japan's forests. Considering high costs often connected to transportation and other logistical costs of woody biomass businesses, these businesses are often greatly dependent on supply chain design (Keefe et al., 2014). In addition, as noted by Yoshida et al. (2014), a key success factor for woody biomass in Japan is the development of rural infrastructure to for harvesting and transportation of wood from the forest. However, opinions on how to address this issue vary. Company A, for instance, builds own road networks, even in areas with steeper terrain, and notes the lack of knowledge on the building of such networks in Japan. Company B, on the other hand, focuses on advanced machinery for harvesting and collection instead of investing in extensive road networks.

4.1.2. Business model integration

Success factor 3), business model integration involves the incorporation of several value pathways in the business model of the company. For example, by utilizing both electricity and heat generated in the power plants, a broader base of value is derived.

Table 2
Comparative analysis of stakeholder success factor perceptions, based on emphasis on the factors during the interviews. S: Strong, M: Moderate, W: Weak.

	(1) Respect of values & traditions	(2) Transportation infrastructure	(3) Business model integration	(4) Relationship & trust	(5) Local vitalization	(6) Biomass quality control
Company A	S	S	M	M	M	W
Company B	S	M	S	M	S	W
Company C	W	S	W	S	W	S
Company D	W	M	M	S	S	S

This is seen as moderately important by Company A, considering plans to integrate biofuel and power production in current business. Company B strongly highlights the importance of integrating biofuel production with the manufacturing of other forest products, for maximization of the value derivable from the tree. Biofuel is created from residual products, and is the last step in the value chain, integrating a diversity of value streams in the company's overall business model. Leban et al. (2016) similarly emphasize the multi-dimensional value to be obtained in contemporary forest management, with woody biomass being a by-product of other forest products and services. In addition, Company C's only source of demand is connected to supplying pellets to a large-scale local power utility. Hence, there is no focus on business model integration. Company D supplies residual heat from power production to local farmers, for strawberry cultivation in greenhouses. This is provided free of charge, but is a clear example of value creation through integration. Company D also creates products from the ash produced during biomass combustion.

4.1.3. Relationships and trust

The establishment of relationships and trust, success factor 4), is seen as a success factor by all stakeholders considering the interdependencies in the woody biomass supply chain for the reaching of objectives of individual stakeholder business models. There is slightly more emphasis on this factor by Company C and D, downstream in the supply chain. This could be explained considering that downstream actors are more dependent on ensuring long-term upstream relationships for a stable supply of wood as biofuel in order to operate. Wang and Watanabe (2016) similarly find the importance of trust-building for long-term cooperation of upstream and downstream stakeholders in the biomass supply chain, with decreased risk perceptions by the upstream partners.

4.1.4. Local vitalization

Local vitalization, success factor 5), is seen as important be companies A, B and D. This is particularly highlighted by Companies B and D. Companies A and B discuss the importance of forestry to enable new job creation in local areas and attract younger people to work in less urban locations. These companies also discuss this in connection to the need to revitalize the forest industry in itself. Company D offers land and residual heat from power production processes to local farmers, also a clear example of engaging in activities aiming to “give back to society” and give life to local industry. As highlighted by Yagi and Nakata (2011), Kishita et al. (2017) and Malico et al. (2016), local woody biomass supply chains can aid in enhancing competitiveness of local industry and contribute to rural socioeconomic development. However, in doing so and for successfully creating value on the long term, effective forest management practices (Malico et al., 2016) and rural infrastructure (Yoshida et al., 2014) are vital elements to be identified and developed.

4.2. Divergence of perceptions

In contrast to the convergences seen, there is a clear divergence of emphasis in perceptions related to the success factors 1) respect of values and traditions and 2) biomass quality control. This indicates an imbalance between the upstream and downstream supply chain stakeholders. Long-term success may involve a variety of factors for individual supply chain players, from the firm's financial growth to expansion. From the supply chain perspective, long-term success refers to the ability to maintain a stable resource flow (Wang and Watanabe, 2016) as well as profitability (Hugos, 2011) from the upstream to downstream, throughout a period decided upon collectively by all stakeholders. Relationship management and coordination are key elements in this (Johansen and Røyrvik, 2014; Wang and Watanabe, 2016), and therefore, understanding and closing the perceptions gap identified in this study may be key to consider for long-term success of the supply chain.

4.2.1. Respect of values and traditions

Success factor 1), respect of values and traditions, involves awareness and consideration of forest-owner history and culture in Japan. Companies in the upstream supply chain, forest-owners, strongly emphasize respect of values and traditions in the forest industry as a main success factor for the entire supply chain. This factor is not emphasized by the downstream stakeholders. Company B indicates that the entire supply chain should be based on sustainable forestry in order to show respect to the forest and forest-owners. Both Company A and Company B discuss the culture associated to forests in Japan, majorly as the forests have been inherited throughout generations.

Forest-owners, Company A and B, also clearly point out that the lack of respect of this factor by other upstream stakeholders, such as electricity producers, is an issue that needs to be managed. If downstream stakeholders obtain an improved understanding of the values and objectives of the forest owners, a foundation for improved communication and integrated supply chain design can be facilitated. Respect of values and traditions is likely to enable building more sustainable long term relationships based on mutual trust and understanding, which will also facilitate increased discussion and collaboration regarding other issues. This is also in line with discussions on social and economic aspects of wood by Aguilar (2014), concerning the “social availability” of wood influenced by different worldviews and value systems, and Keefe et al. (2014), with regard to management objectives among forest owners which are important to be considered in planning the woody biomass supply chain. The importance of managing values of all stakeholders for long-term success is also highlighted by White and Bourne (2007).

4.2.2. Biomass quality and control

Success factor 6), biomass quality and control involves procedures for biomass quality assurance, such as moisture content, ash content and size. This factor is strongly highlighted downstream in the supply chain by Companies C and D, i.e. biofuel and electricity/heat producers. Biomass quality is important for successful power generation, explaining its increased emphasis in the downstream supply chain and little emphasis upstream. Improved management of biomass quality, especially moisture content, in the upstream woody biomass supply chain can aid in improving energy conversion efficiency and cost-effectiveness across the supply chain (Kishita et al., 2017; Greene et al., 2014; Malek et al., 2017).

In tackling the issue of biomass quality management, increased awareness, co-planning and knowledge-sharing could enable supply chain effectiveness. Similarly indicated by de la Fuente et al. (2017), by moving woody biomass production processes and knowledge further upstream in the supply chain, higher volumes and quality of woody biomass is obtainable. An example of this knowledge-sharing is to share that it might be more efficient, economically and logistically, to move some of activities related to woody biomass drying processes carried out in the downstream to the upstream stakeholders. As indicated by BR&D (2014), the holistic efficiency of the woody biomass supply chain can be improved by drying forest residues in the field prior to transportation, for an increased quality of material flow. For instance, leaving the wood to dry in the forest can reduce costs of transportation due to lighter weights at the same time as reducing the need for drying in the later stages, requiring space and additional energy costs from

downstream actors. As mentioned by Company B, available space is an issue for many companies with regard to woody biomass drying. The interviewee discusses that it would be more economical to stock the logs for a longer time, leaving them to dry in piles outdoors after receiving them. This would cut the costs of operating drying machines, but land price is an issue. Forest owners, however, already own the land, and one technique efficient for drying wood is leaving logs in the forest with smaller branches and leaves on, which will continue to absorb the water for diffusion into the air even after the tree has been cut down (Kamijo et al., Personal communication). An open discussion regarding this issue, that might not be very familiar to forest owners that have little experience of supplying wood for energy purposes, may consequently open up for new, more efficient configurations of the required biomass moisture control, drying and quality management processes across the supply chain.

4.3. Policy implications for understanding and managing perceptions in the woody biomass supply chain

It can be suggested that understanding the perceptions of multiple stakeholders as well as convergences and divergences among them is of importance in improving policy creation processes, if these policies thereby can more effectively address reality. As expressed by Shiroyama et al. (2015), finding interests and challenges in complex, diverse stakeholder networks is vital in enabling consensus and project feasibility. Understanding perspectives and social dynamics is important to be increasingly included in policy-making for sociotechnical systems (Bolton and Foxon, 2015), such as woody biomass supply chains. It is suggested that by managing divergent perceptions, and closing the perceptions gap seen in this study, a more effective supply chain can be enabled. It is of interest to explore how to leverage convergences and manage the divergences seen in this paper through policy measures. It is found that success factors highlighted incorporate both formal factors, such as biomass quality control and transportation infrastructure, as well as informal factors, such as relationships and trust. It can therefore be suggested that it is valuable to incorporate formal as well as informal elements in policy-making, considering the significance of both dimensions in analyses of sociotechnical systems with complex stakeholder networks. The limited scope of this study, involving four interviewed companies, cannot alone lay a basis for universal policy recommendations. However, it is suggested that the findings of this paper can provide a direction for policy research integrating both formal and informal factors affecting the woody biomass supply chain. With this, policy implications of this study are suggested as follows, targeting elements across the woody biomass supply chain.

4.3.1. Cross-ministerial integration

By enhanced collaboration and policy integration on the government level, it is suggested that strategies and incentives pertaining to woody biomass can be more coherent on a holistic level in industry. This can be done, for instance in the case of Japan, through the increased integration of and communication between the Forest Agency of the Ministry of Agriculture, Forestry and Fisheries (MAFF) and energy-related agencies in METI, such as the Agency for Natural Resources and Energy. Kishita et al. (2017) and Malico et al. (2016) highlight the ability of woody biomass business to contribute to development in terms of socioeconomic progress, energy security and emissions reduction. It can be highlighted that these businesses are connected to multiple governmental goals, such as climate change mitigation, local socioeconomic vitalization, improved forestry and energy security. It is therefore suggested that sustainable forest strategies can be enabled by energy strategies, and vice versa, for stronger interconnection and coherence from the upstream to downstream woody biomass supply chain. It is suggested that both formal and informal challenges stretching across the supply chain, such as a lack of understanding of forestry culture as well as of biomass quality control, can be addressed through this integrated approach. Through an integrated approach in policy, it is suggested that multiple elements of value can be achieved from the forest and across the woody biomass supply chain, leading to heightened attainment of governmental goals, such as local society revitalization, increased shares of renewable energy and emissions reduction.

4.3.2. Knowledge-building on wood logistics

There a general unison in the case studies of this paper that woody biomass transportation infrastructure necessitates improvement in Japan, also as discussed by Yoshida et al. (2014). In addition, enhanced educational levels in the upstream biomass supply chain may be a key factor for decreased risk perceptions and successful, long-term cooperation between upstream and downstream stakeholders (Wang and Watanabe, 2016). With this, it is suggested that knowledge-building through local training programs on wood logistics, from harvesting to collection to transportation from the forest, is of interest to be explored further for Japan. The aspects of sustainable, productive forestry as well as of biomass quality control are suggested to be included in these programs, especially as these are main challenges recognized. Knowledge management can in turn aid in building trust, decreasing risk perceptions and thereby strengthening relationships based on mutual understanding across the supply chain.

4.3.3. Forest certification

As discussed by Gain and Watanabe (2014), lacking documentation and forest certification has been one of the causes behind unproductive forestry in Japan. Through improved certification of forests and forest products which would be utilizable in business contexts, it is suggested that a higher level of trust and lower risk perceptions can be attained in supply chains, both pertaining to woody biomass as well as other forest-derived products. The importance of trust-building for reduced risk perceptions for long-term cooperation is also discussed by Wang and Watanabe (2016). It is suggested that certification improvement may enable improved contracts between the forest, biomass and energy industries, thereby increasing transparency, supporting biomass quality control and aiding in the building of stronger relationships and trust, identified as a success factor in this study.

4.3.4. Local coordinator

In this paper, innovation in the woody biomass supply chain is defined, in line with Hartmann (2014), as the creation of new value through new technologies, processes or organizational structures, by one or more organizations. An example is the potential added value of the mobile wood-chipping process being investigated by Company A. Hugos (2011) and Johansen and Røyrvik (2014) discuss the ability of supply chain stakeholder coordination to contribute to profitability on the long-term as well as to enable innovation. The understanding of multiple perspectives and mindfulness concerning working relations in a supply chain in turn aids openness and the collective creation of new value (Hugos, 2011). Considering the importance of stakeholder coordination and understanding for innovation, it is suggested that through the inclusion of a coordinative body in the woody biomass supply chain to link efforts, enable knowledge-sharing and spread awareness, the perceptions divergence can more effectively be managed. This body is meant to have a purely coordinative role, catalyze discussion for relationship-building, resolve conflict and enable informed

decision-making. A coordinator could especially enable the informal success factors identified in this study, including respect of values and traditions, and relationships and trust. This may in turn support formal factors, such as planning for improved biomass quality control across the supply chain, or finding new pathways for value creation through business model integration.

4.3.5. Biomass quality control standards

There is a multitude of research showing the importance of biomass quality management from the upstream woody biomass supply chain for a more effective material flow and improved energy conversion processes in the downstream (Kishita et al., 2017; Greene et al., 2014; Malek et al., 2017). With this, for an increased understanding and routine management of biomass quality, as well as communication surrounding terms of biomass quality across the supply chain, it is suggested that biomass quality standards and procedures are of interest to explore in the realm of policy and national directives.

4.3.6. Feed-in-tariff (FIT) for heat

Yagi and Nakata (2011) highlight the ability of biomass operations for power and heat to enhance the competitiveness of local industries. The Japanese government similarly discusses the importance of generating both power and heat from woody biomass in order to reach higher economic efficiencies and to base the business on local conditions and demands (METI, 2014). With this, and considering the convergence of opinions on the significance of business model integration and local vitalization, it is suggested that a model for a FIT for heat from renewable sources is of interested to investigate further in the context of Japan. This may increase local integrations of new, small-scale woody biomass energy businesses, thereby vitalizing local economies. This would in turn be in line with the Japanese government's local community revitalization efforts.

4.3.7. Formal and informal dimensions: reinforcing effects

Formal aspects in the given policy indications, such as certification, wood logistics training programs, an official coordinator, governmental structural change, FITs for heat and biomass quality standards may aid in laying a ground upon which woody biomass supply chains can stabilize and grow in Japan. Related informal aspects, such as knowledge-sharing, relationship- and trust-building, improved communication and cross-ministerial collaboration may aid in enabling higher policy impact and sustainable growth on the long-term. It is suggested that both informal and formal dimensions are vital in policy-making for sociotechnical systems, as the integration of technology and social dynamics creates opportunities and challenges in both spectrums.

5. Conclusions

Despite an abundance of forest alongside woody biomass-related policies and financial incentives in Japan, challenges are still being faced by the bio-power and heat industry. With this, the importance of understanding and managing convergences and divergences of perceptions among stakeholders in the woody biomass supply chain in the context of policy-making was explored in this paper. Based on perspectives from stakeholders from across the supply chain, a series of success factors were identified and can be seen as formal (transportation infrastructure, business model integration and biomass quality control), as well as informal (respect of values & traditions, relationship & trust and local vitalization). It is important to consider that relevant success factors are contingent upon local conditions and stakeholder synergies of different projects, and that the factors found in this study may not be universally applicable. This would require further research to establish. However, it is suggested that a convergence as well as divergence of perceptions, or a perceptions gap, among upstream and downstream stakeholders is important to be bridged in order to further stimulate locally sourced woody biomass businesses. The existence of this divergence implies the importance of approaching woody biomass energy system development with a supply chain perspective. Moreover, as it is found that challenges and success factors of the woody biomass supply chain are traceable to both formal and informal dimensions, it can be suggested that woody biomass supply chains should be planned and managed from both dimensions. A series of policy indications were suggested based on this reasoning, including cross-ministerial integration, knowledge-building on wood logistics, forest certification, a local coordinator, biomass quality control standards and a feed-in-tariff for heat. The convergence and divergence of perceptions as well as the influence of both informal and formal factors across the woody biomass supply chain as found in this paper suggest vital considerations in policy-making. This can be a significant implication for energy policy in Japan and globally.

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